PROJECT HANNIBAL // DAILY BRIEF

Fri 23 Oct 2026, 21:03 UTC · receive-only · sources NASA/JPL, SatNOGS, CelesTrak

No change since last report. Apophis EXCLUDED (orbit 220, Lisbon IMPOSSIBLE, JPL vs ESA CONSISTENT); satellites 6/7 fresh; 5 debris shells over the line.

No change since the last report (2026-10-23_2039).

1 Apophis (99942): where it won't be, where it will be

Verdict EXCLUDED. Orbit solution 220 (U=0, 7,370 observations, last 2022-04-09). Sentry: removed 2021-02-21 08:22:28.

pass (TDB)	distance km	nearest edge km	window km	above atmos. km	verdict
2029-Apr-13 21:46	38,011	38,008	6.6	31,530	EXCLUDED
2051-Apr-20 02:05	6,206,849	5,969,533	474,687.9	5,963,055	EXCLUDED
2102-Sep-11 20:52	3,499,720	3,365,402	491,560.6	3,358,924	EXCLUDED
2116-Apr-12 21:44	2,627,073	154,849	15,452,640.3	148,371	EXCLUDED ~rough

2029 closest at 2029-Apr-13 21:45 UTC: 38,015 km from Earth's centre, over 29.0N 43.1W; 3-sigma box 2.5 × 1.7 km, ±3.3 km in range.

Impossible-event test, impact off Lisbon: IMPOSSIBLE. Closest 32,495 km at 2029-Apr-13 21:39 UTC, nearest 3-sigma edge 32,492 km (32,392 km above the atmosphere). Naked-eye from Lisbon from 19:41 UTC, up to mag 2.8.

envelope (TDB)	along km	across km	± range km	time ±
2029-Apr-13 21:46	2.5	1.7	3.3	< 00:01
2051-Apr-20 02:05	221,279.6	49.4	237,314.3	08:07
2102-Sep-11 20:52	1,887,133.1	206.7	243,130.7	15:17
2116-Apr-12 21:44	10,290,190.5	24.7	12,716,358.3	7_11:21 ~rough

1b Independent check: NASA/JPL vs ESA NEOCC (pre-registered criteria C1-C5)

CONSISTENT: JPL and ESA agree within their uncertainties, no pass reaches the atmosphere in either, and neither risk list carries the object.

Risk lists: JPL Sentry removed 2021-02-21 08:22:28; ESA not listed. JPL orbit 220 (7370 obs used, last 2022-04-09); ESA file 9577 obs, last 2022-04-09.

pass (JPL TDB)	JPL km	ESA km	diff km	sigma	status
2029-Apr-13 21:46	38,011	38,014	+2.3	1.4	AGREE
2044-Aug-24 07:17	12,012,181	12,175,047	+162,866.3	1.6	AGREE
2051-Apr-20 02:05	6,206,849	6,009,817	-197,031.6	1.6	AGREE
2066-Sep-16 01:59	10,387,038	11,142,091	+755,052.8	1.5	AGREE
2073-Jul-27 18:10	16,797,951	16,594,191	-203,759.6	1.5	AGREE
2080-May-09 07:59	13,002,041	13,154,451	+152,410.6	1.4	AGREE
2087-Apr-07 12:20	14,326,096	14,534,893	+208,797.0	1.6	AGREE
2102-Sep-11 20:52	3,499,720	3,803,329	+303,608.4	1.4	AGREE
2116-Apr-12 21:44	2,627,073	12,161,757	+9,534,684.0	2.0	AGREE
2109-Mar-18 20:01 UTC	-	27,279,410			ESA only
2109-Jun-11 06:16 UTC	-	18,108,300			ESA only
2124-Sep-21 03:37 UTC	-	22,147,821			ESA only

Evidence: 6/6 sources fetched live, raw replies hashed in docs/investigation/apophis/evidence/2026-10-23_2103 (manifest SHA-256 a51f86d05e4b078d...).

2 Satellite health (SatNOGS, receive-only)

satellite	NORAD	state	last good (UTC)	age h	stations*	next pass (UTC)	DB
ISS (ZARYA)	25544	LIVE	2026-10-23 19:51	1	20/25	10-25 18:10 SGP	in orbit
FUNcube-1 (AO-73)	39444	LIVE	2026-10-23 19:39	1	9/25	10-24 19:20 JF6FYI Nishikazu,	in orbit
ESTCube-1	39161	SILENT	2026-08-18 11:26	1594	17/25	-	in orbit
LAPAN-A2 (IO-86)	40931	LIVE	2026-10-23 20:17	1	5/25	10-25 05:22 PAPUA OBSERVER	in orbit
METEOR M-2	40069	LIVE	2026-10-22 08:52	36	11/25	10-23 21:22 Test	in orbit
OSCAR 7 (AO-7)	7530	LIVE	2026-10-23 09:58	11	7/25	10-24 21:42 biGS	in orbit
JAS-2 (FO-29)	24278	LIVE	2026-10-23 14:19	7	9/25	10-24 20:16 biGS	in orbit

LIVE = a good decode within 30 days; stale = none in 48 h. *distinct stations among the latest good observations (SatNOGS pages cap at 25).

3 Debris cascade watch (CelesTrak SATCAT)

In orbit 35,070; working satellites 17,282; debris 12,539; rocket stages 2,294. Shells over the line: 1150-1200, 1400-1450, 1450-1500, 1500-1550, 1550-1600 km.

shell km	objects	cascade index	band
1400-1450	459	1.82	OVER THE LINE
1450-1500	500	1.54	OVER THE LINE
1550-1600	224	1.34	OVER THE LINE
1150-1200	577	1.22	OVER THE LINE
1500-1550	271	1.18	OVER THE LINE

Cascade index = objects / linearised Kessler threshold; a comparison between altitudes, never a forecast.

Generated by the Hannibal daily routine. Receive-only monitoring: nothing was transmitted. Sections that could not reach their source say NO DATA.